Condition Fidelity Under Corruption: A Robustness Study of ControlNet

Isabella Wu Nicole Xu
Massachusetts Institute of Technology
{isawu888,nicolexu}@mit.edu

Abstract

Controllable text-to-image generation methods, such as ControlNet, enable spatially precise image synthesis by conditioning a diffusion model on structural user inputs like edge or depth maps. However, despite the widespread adoption of such methods, condition fidelity remains poorly characterized: existing evaluations are largely qualitative, and robustness to noisy or incomplete conditions is rarely measured. We present two main contributions toward closing these gaps.

First, training ControlNet on a Stable Diffusion 1.5 backbone using the Fill50K circle-contour dataset, we introduce a quantitative condition fidelity evaluation framework using two complementary metrics: Canny-based edge detection and RANSAC-based geometric estimation. We also evaluate prompt fidelity and image quality using CLIP score and FID. Second, we conduct a controlled input corruption study, evaluating three different classes of condition perturbations to characterize how condition following degrades under imperfect inputs.

On clean inputs, our model achieves a Circle IoU of 0.925 and Canny F1 of 0.637, with CLIP score 30.26 and FID 58.79. Under corruption, prompt fidelity and perceptual quality remain stable while condition fidelity degrades, exposing a fidelity-quality gap that standard metrics fail to detect. Our methodology and results reveal that condition fidelity is both measurable and sensitive to realistic input corruptions, highlighting the need for explicit condition fidelity evaluation and multi-axis assessment of controllable generation systems.

1. Introduction

Just how much control do we have over generative image models? Text-to-image (T2I) diffusion models have become a major focus in both computer vision and the broader public, due to their ability to generate high-quality, photorealistic images from simple natural language prompts. Models such as Stable Diffusion [9] produce semantically relevant and visually compelling outputs at a relatively low computational cost.

However, text alone is often too weak to specify precise spatial structure. Users typically have a clear mental image of their desired image composition, such as where certain edges should lie or how figures should be arranged. Oftentimes, a T2I model's output may not match the user's mental image, especially in scenes that contain complex layouts or poses. As a result, users are often forced to go through a trial-and-error process of repeatedly editing a prompt and inspecting the generated result until it is satisfactory. This has real consequences: in applications where spatial structure matters, including architectural design or medical image synthesis, a model that ignores its conditioning input is not solving the intended problem, and standard quality metrics may not detect this failure.

ControlNet addressed this problem by introducing a trainable neural network module that conditions a frozen T2I diffusion backbone [12] on a user-provided *condition* image containing structural or semantic guidance, which can come in the form of a Canny edge map, depth map, segmentation mask, user scribble, etc. Notably, [12] demonstrated that strong spatial controls could be added to a T2I without retraining the full diffusion model or damaging the large-scale pretrained backbone.

Since then, several works have explored related directions. T2I-Adapter injects control signals via lightweight adapter modules while keeping the original model frozen, which allows for efficient training and flexible integration [6]. UniControl unifies multiple control modes within a single framework rather than training a separate model for each condition [7]. Overall, these methods extend controllable generation to more conditions or improve efficiency.

A main challenge now is not whether controllable generation is possible, but how faithfully a model can follow its control input. More recent works such as ControlNet++ [5] highlight existing methods still do not explicitly optimize *condition fidelity*——how closely a generated image adheres to the conditioning input. In some cases, ControlNet may generate high-quality images that are visually plausible but do not necessarily adhere to the intended condition. The authors of ControlNet++ propose a reward model that optimizes a "consistency loss" between the input condition and the condition extracted from the generated image [5].

Still, this line of work primarily focuses on improving consistency of images generated under clean, well-formed conditions (*e.g.*, reference images or artist-provided). In practice, condition inputs are frequently noisy, incomplete, or out-of-distribution, especially when provided by non-expert users [2]. This can potentially lead to unexpected or undesirable outputs and hinder users from generating satisfactory images [11]. While Xuan et al. [11] study ControlNet behavior under inexplicit masks and propose an architectural fix, no prior work systematically characterizes how condition fidelity degrades as a function of corruption type and severity in controllable text-to-image generation. This gap motivates our work.

The central question we aim to answer is: how robustly does a trained ControlNet preserve condition fidelity when the conditioning input is corrupted? While ControlNet [12], T2I-Adapter [6], and UniControl [7] show that structured control can be incorporated into diffusion generation, they do not center robustness to imperfect conditions as the main object of study. In contrast, this project focuses on evaluation and characterization. To enable precise condition fidelity measurement, we train a ControlNet on a Stable Diffusion 1.5 backbone using the Fill50K dataset [12], which consists of circle-contour conditioning inputs whose geometric simplicity allows exact measurement of structural adherence. We evaluate performance along three axes: *prompt fidelity* (how closely a generated image adheres to the text prompt), *image quality* (perceptual similarity to the ground truth image), and *condition fidelity*, measured via two complementary metrics, Canny-based edge comparison and RANSAC-based geometric estimation, that together capture distinct structural failure modes for our circle-contour condition. We then introduce controlled corruptions to the conditioning inputs, including arc occlusion, edge jitter, and Gaussian blur at four severity levels, and measure how each metric degrades, holding all other variables fixed.

Our main contributions are threefold. First, we introduce a rigorous condition fidelity evaluation framework using two complementary metrics, Canny-based edge comparison and RANSAC-based geometric estimation, that together capture distinct failure modes for circle-contour conditions, going beyond the single-extractor approach of prior work [5]. Second, we conduct a controlled corruption study evaluating three perturbation types (arc occlusion, edge jitter, Gaussian blur) at four severity levels, characterizing how condition fidelity degrades under imperfect inputs. Third, we identify and characterize a *fidelity-quality gap*: standard image quality and prompt fidelity metrics fail to detect condition fidelity failures, meaning a model can appear to generate images visually similar to the prompt or original image, but fail at well-following the condition input. By characterizing how and when condition fidelity fails, this work aims to provide a foundation for more rigorous evaluation of controllable generation systems.

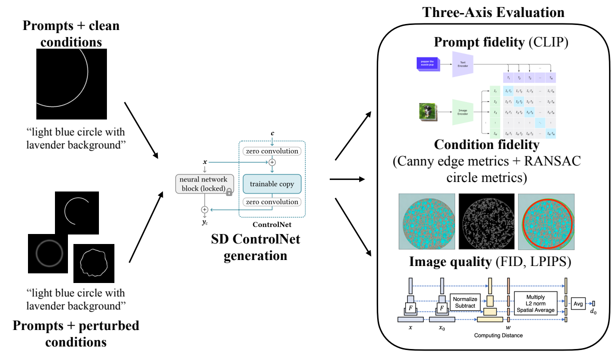

Figure 1. **Overview of the evaluation pipeline.** We evaluate ControlNet generation under both clean and perturbed conditioning inputs. Given a text prompt and either the original circle contour or a corrupted version of the contour, the trained SD-ControlNet model generates an output image. Each generated image is then evaluated along three axes: prompt fidelity using CLIP score, image quality using FID and LPIPS, and condition fidelity using Canny-based edge metrics and RANSAC-based circle estimation. This pipeline allows us to measure whether generated images remain visually plausible and prompt-aligned while accurately following the conditioning input.

2. Methods

Our goal is to characterize how well a trained ControlNet follows its conditioning input, both under clean conditions and under realistic input corruptions.

To this end, we design an evaluation framework (Figure 1) that measures condition fidelity precisely alongside prompt fidelity and image quality, and construct a controlled corruption protocol that varies perturbation type and severity independently. We first describe our dataset, which enables precise geometric measurements due to its simplicity. Next, we outline our training setup, evaluation metrics, and corruption protocol. Our code, trained checkpoints, and evaluation scripts are available at `https://github.com/nicolexxuu/6s058-final-project-controlnet`.

2.1. Dataset and Splits

We train and evaluate our model using the Fill50K dataset [12], which contains 50,000 image-condition-prompt triples. Each triple consists of a target image of a colored filled circle on a solid background, a binary contour map as the conditioning input, and a natural language prompt describing the colors of the circle and background.

We choose Fill50K because a circle is fully characterized by its center, radius, and color, enabling exact condition fidelity measurement rather than approximation, unlike complex conditions such as human poses or depth maps where correct condition following is more subjective. This precision enables the circle-specific geometric metrics we introduce in Section 2.4. Fill50K is also the official ControlNet tutorial dataset, making our setup consistent with a standard reference implementation.

We partition the dataset with a 80/10/10 train/val/test split using a fixed random seed, yielding 40,000 training images and 5,000 test images, large enough for stable metric estimation.

2.2. Model Architecture and Training

We train a standard ControlNet [12] on a frozen Stable Diffusion 1.5 backbone [9] without modification, using SD v1.5 to match the backbone used in both ControlNet and ControlNet++ [5]. Since our focus is robustness evaluation, we avoid architectural changes that would complicate comparison with prior work. The SD backbone is kept frozen (`sd_locked=True`) so that condition fidelity can be attributed to the ControlNet module rather than the generative backbone, and the full encoder is trained rather than the mid-block only (`only_mid_control=False`).

We train for 2 epochs (11k training steps) with batch size 8, learning rate 1×10^{-5}, and float32 precision, following the official ControlNet training documentation [12], on an NVIDIA A100 GPU.

2.3. Inference

Given the geometric simplicity of the dataset, images were generated using a DDIM sampler [10] with 20 denoising steps, classifier-free guidance scale 9.0, and $\eta = 0.0$ at 512×512 resolution. These settings are fixed across all experiments so that any performance differences reflect changes in the conditioning input alone.

2.4. Evaluation Metrics

We evaluate generated images along three axes: prompt fidelity, image quality, and condition fidelity. For prompt fidelity and image quality, we adopt CLIP score [8] and FID [4] respectively, following the evaluation protocol of ControlNet [12] and ControlNet++ [5]. Our primary contribution lies in condition fidelity, where we introduce two complementary metric families, Canny-based edge-level comparison and RANSAC-based geometric estimation, that together enable a more precise characterization of condition fidelity than prior work provides.

Prompt Fidelity. We measure prompt fidelity using CLIP score [8], which computes cosine similarity between image and text prompt embeddings. Fill50K's short descriptive prompts (e.g., "a blue circle on a brown background"), make CLIP an appropriate measure of color and semantic alignment.

Image Quality. We measure image quality using two complementary metrics. Fréchet Inception Distance (FID) [4] measures the distributional distance between the full set of 5,000 generated test images and their corresponding ground truth images, capturing whether their distributions are statistically similar. Learned Perceptual Image Patch Similarity (LPIPS) [13] complements FID by providing a per-image perceptual distance from ground truth, enabling image quality to be analyzed alongside per-image condition fidelity metrics. A key motivation for including FID and LPIPS alongside condition fidelity metrics is to detect what we call the *fidelity-quality gap*: a model may continue generating visually realistic images even as it loses adherence to the conditioning input. This failure mode is particularly insidious in practice, as high image quality can mask poor condition following.

Condition Fidelity. A key limitation of the original ControlNet evaluation is that condition fidelity is assessed primarily through qualitative user studies, with quantitative metrics provided only for the segmentation condition [12]. Notably, no quantitative condition fidelity metrics are provided for edge-based conditions specifically. ControlNet++ addresses this by introducing a cycle consistency framework: rather than comparing the generated image to the condition directly, it extracts the same type of structural representation from the generated image and compares it to the original condition. However, ControlNet++ applies this idea using a single extractor per condition type, which captures only one aspect of adherence. For circle-contour conditions, edge appearance and geometric position can fail independently, motivating two complementary extractors that together provide a more complete characterization than either prior work achieves.

Inspired by this framework, we adopt cycle consistency evaluation but extend it with two complementary extractors suited to our circle condition. A naive pixel-level comparison is not meaningful here, as the condition is a binary contour on a black background, while the generated image is a colored filled circle on a textured background. Even a perfect generation would show near-zero pixel-level similarity to the condition. Instead, we extract structural information from generated images in two ways that capture geometrically distinct failure modes, described below.

Edge-based metrics. We extract edges from each generated image using the Canny edge detector [1] and compare the resulting edge map against the original binary condition. We report IoU, F1 score (with a 3-pixel tolerance to account for minor edge thickness variation), and Chamfer distance (average nearest-neighbor distance between edge

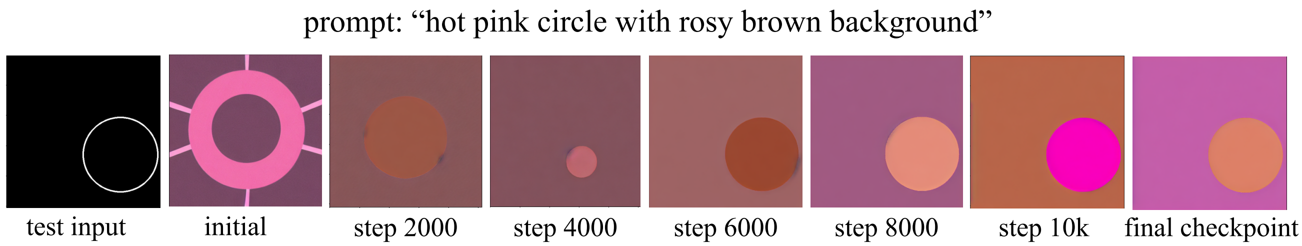

Figure 2. **Training ControlNet on Fill50K**. Because ControlNet initializes a trainable copy of the SD encoder with zero convolutions, ControlNet always predicts plausible, high-quality images during the entire training process. At around 6000 training steps, the model learns to follow the input condition. However, throughout the entire training process, foreground and background colors are occasionally swapped or blended, indicating that the semantic binding between prompt attributes and image regions has not been fully learned. This suggests condition fidelity does not guarantee prompt fidelity.

point sets), providing a continuous spatial proximity measure even when edge maps do not overlap. IoU and F1 are both bounded in $[0, 1]$ but differ in strictness: IoU requires exact pixel overlap, while F1 is computed with a 3-pixel tolerance (a detected edge counts as correct if it falls within 3 pixels of any condition edge), accommodating the thickness difference between the thin condition contour and the typically thicker boundary of generated circles. These metrics cover 100% of the test set, including images where the condition circle extends beyond the image boundary. However, Canny-based metrics are sensitive to edge texture, rendering artifacts, and boundary thickness rather than geometric position. A circle with rough or thick edges will score poorly even if it is geometrically correct.

Geometry-based metrics. To complement pixel-level comparison, we apply RANSAC-based circle fitting [3]. We extract Canny edges from the generated image and fit the parameters of a circle, (c_x, c_y, r) to the edge map. We iteratively sample edge point subsets, estimate (x, y, r), and select the hypothesis with the largest inlier set (edge points within a fixed distance threshold ϵ of the fitted circle. This makes RANSAC tolerant to noisy or partially occluded edges.

A detection is considered successful if RANSAC finds a circle with at least a minimum inlier threshold, and images where detection fails are excluded from geometric metrics. For generated images on which RANSAC detects a circle, we compare the estimated (x, y, r) directly to those of the condition circle, reporting normalized center error, normalized radius error (both normalized by condition radius to be scale-invariant), and circle Intersection-over-Union (IoU).

Canny metrics detect failures in edge appearance but may miss geometric misplacement. Geometric metrics immediately detect position and size errors but are insensitive to edge quality. For instance, a generated circle with noisy edges in the correct location will score poorly on Canny IoU but perfectly on RANSAC center error. Conversely, a smooth-edged circle shifted from the condition location will score moderately on Canny IoU but poorly on RANSAC center error. Together, both classes of metrics provide a more complete characterization of condition fidelity than either alone, and a stronger evaluation than the single-extractor approach of ControlNet++.

2.5. Corruption Protocol

While prior work including ControlNet [12] and ControlNet++ [5] evaluate condition fidelity under clean, well-formed inputs, real-world conditioning inputs are often imperfect. For example, users may provide incomplete sketches or poorly extracted edge maps. Relatively less attention has been paid to systematically characterize how condition fidelity degrades as a function of input corruption type and severity. We address this gap by introducing a controlled corruption study as a main contribution of this paper.

We perturb the conditioning inputs of a 300-image subset of the test set (Figure 3). This subset keeps generation costs tractable across all 12 corruption conditions (3 types x 4 severities) while remaining large enough for stable metric estimation. We choose three different types of perturbations to represent different deviations that may often occur in real-world edge map conditions. We perform the perturbation on each clean condition at four increasing severity levels (0.1, 0.2, 0.4, 0.6), and evaluate them using the same metrics defined in Section 2.4. All metrics are computed against the original clean condition rather than the corrupted one, so that scores reflect how well the model preserved the intended structure despite receiving a degraded input. Next, we introduce the 3 perturbation types used.

Arc occlusion. A randomly selected contiguous arc segment is removed from the circle contour, with the removed fraction increasing with severity. This tests whether the model can complete or infer a shape from partially missing structural information.

Edge blur. A Gaussian blur kernel is applied to the condition, and the kernel width increases with corruption intensity. This corruption tests whether the model can produce crisp images given a modified edge conditioning signal.

Edge jitter. Edges are randomly jittered, producing an image similar to a handdrawn or imperfectly extracted contour where the user-intended structure is present but partially distorted. For this experiment, the jitter is produced by perturbing contour pixels radially with smoothed random offsets. This corruption tests whether the model is robust to local geometric noise.

2.6. Individual Contributions

Both partners contributed equally to project formulation, literature review, and the writing of this report. For training, both partners collaborated on the training pipeline setup. Isabella worked on the data preprocessing and loading the diffusion model architecture and framework, and Nicole led experiments on training convergence and checkpoint analysis. For evaluation, both partners collaborated on the evaluation framework design. Isabella implemented the Canny-based edge metrics and FID/LPIPS computation, and Nicole implemented RANSAC-based geometric metrics and CLIP score evaluation. For the corruption study, Isabella implemented Gaussian blur corruption and the severity iteration framework, and Nicole implemented arc occlusion and edge jitter corruptions. Both partners contributed to figure design and captions, and to the quantitative and qualitative analyses presented in this paper.

3. Results & Discussion

We present results in three parts. First, we establish a clean baseline by evaluating our trained ControlNet on 5,000 held-out test images across all three axes: prompt fidelity, image quality, and condition fidelity. This baseline directly addresses our central question of how well a trained ControlNet follows its conditioning input, and characterizes the fidelity-quality gap. Second, we present our controlled corruption study that isolates the effect of corruption type and severity on each evaluation axis, holding the model, inference settings, evaluation metrics, and test subset fixed across all conditions. This allows us to directly attribute performance difference to the corruption alone. Third, we analyze metric independence using Pearson correlations between prompt fidelity, image quality, and condition fidelity, isolating the contribution of each and explaining why no single metric is sufficient to characterize controllable generation performance. Together, these experiments reveal that condition fidelity degrades independently of image quality because the model has learned pattern matching rather than geometric reasoning, and that different corruption types expose this limitation to different degrees.

3.1. Training Convergence

Training converged smoothly over 10,000 steps, with final training loss of 0.00508. Qualitative inspection of samples at intermediate checkpoints, pictured in Figure 2 confirms the sudden convergence phenomenon described in [12]: generations at 2k steps show little condition following, while by 6k steps the model reliably produces circles consistent with the conditioning input. All following evaluations use the final model trained with 11k gradient steps.

3.2. Clean Baseline Evaluation

We evaluate our trained ControlNet on 5,000 held-out test images under unperturbed conditioning inputs. Results are summarized in Table 1 and representative generated images are shown in Figure 4. Our evaluation pipeline, illustrating how Canny and RANSAC metrics are computed on generated images, is shown in Figure 5.

Prompt fidelity and image quality. The model achieves a mean CLIP logit score of 30.26 ($\pm$2.91), with a narrow approximately Gaussian distribution (Figure 6), indicating consistent semantic alignment with text prompts across the test set. FID is 58.79, which is moderate relative to state-of-the-art generative models but expected given the limited training duration and batch size. Generated images tend to produce muted colors and flattened backgrounds compared to the vivid, high-contrast ground truth, as visible in Figure 4.

Condition Fidelity: Edge-based metrics. We report mean Canny IoU of 0.161 ($\pm$0.086), mean Canny F1 of 0.637 ($\pm$0.385), and mean Chamfer distance of 19.80px ($\pm$30.36), computed between the binary condition contour and the Canny edge map extracted from the generated image. The substantially higher mean F1 relative to IoU reflects the 3-pixel tolerance accommodating edge thickness difference. Canny exhibits two failure modes in our setting that produce notably low scores. First, generated images with heavy internal texture produce noisy edge maps that score poorly, even when the circle geometry is correct (Figure 5, Row 2). Second, low-contrast generations cause Canny to detect no boundary at all, giving zero IoU and F1 despite a visible circle (Row 3). Chamfer distances follow the same trend, exhibiting a long right tail and high standard deviation ($\pm$32.73).

The distributions of Canny IoU and F1 are notably bimodal (Figure 6): images either produce detectable edges that align with the condition or fail to produce detectable edges entirely, with few intermediate cases. This suggests the model does not degrade linearly. Condition following tends to be all-or-nothing rather than a continuous one, pos-

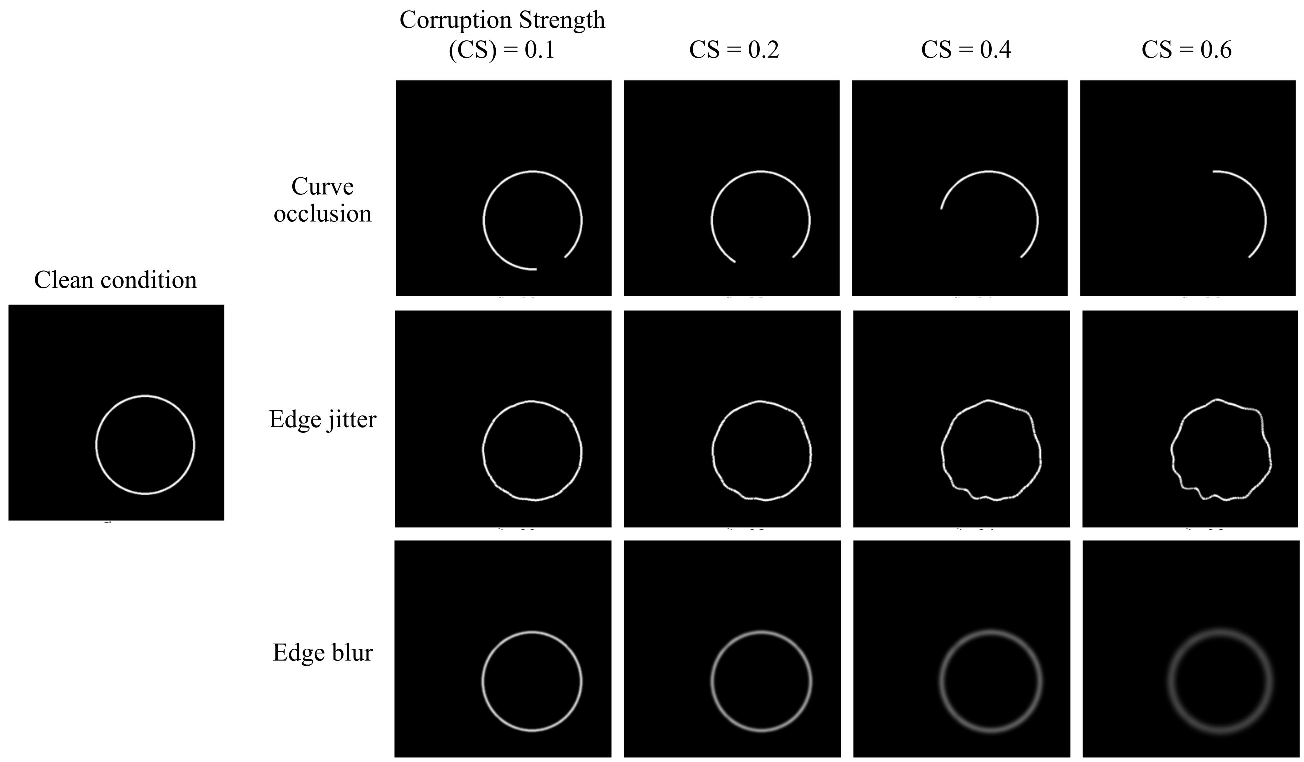

Figure 3. **Conditioning input corruptions at four severity levels.** We evaluate ControlNet robustness under three types of conditioning input corruption at four severity levels (CS=0.1, 0.2, 0.4, 0.6). Row 1 (arc occlusion): increasing arc segments are removed from the circle contour. Row 2 (edge jitter): contour pixels are perturbed radially with smoothed random offsets, simulating a hand-drawn or imperfectly extracted edge map. Row 3 (Gaussian blur): a blur kernel of increasing width is applied to the contour.

sibly reflecting varying prompt difficulty across the test set (colors such as "blanched almond" or "lilac" tend to lead the model into generating textures containing almonds or flowers, for example).

Condition Fidelity: Geometry-based metrics. RANSAC-based circle fitting achieves a circle detection rate of 93.5% across the test set. Among detected images, mean Circle IoU is 0.925 ($\pm$0.157), and heavily concentrated near 1.0 (Figure 6). Mean center error is 7.94px (1.55% of image width) and radius error is 7.71px (1.51% of image width). Both metrics have a moderate variance (std $\approx$ 18px). These results are strong; when the model generates a detectable circle, it is almost always in the right place at roughly the right size.

Fidelity-Quality Gap and Metric Independence. A key observation is that image quality and condition fidelity are partially decoupled. Generated images frequently appear visually plausible while simultaneously scoring poorly on condition fidelity metrics. At the dataset level, FID of 58.79 confirms a distributional gap between generated and real images. At the per-image level, mean LPIPS is 0.524 ($\pm$0.220), indicating moderate perceptual distance from ground truth. Crucially, LPIPS shows only weak correlation with condition fidelity ($r = -0.240$ with Canny IoU, $r = -0.203$ with Circle IoU, $p < 0.001$ in both cases), with $r^2 < 0.06$, meaning image quality explains less than 6% of variance in condition fidelity. A model can generate perceptually realistic images while failing to follow the conditioning input.

To further quantify metric independence, we compute Pearson correlations between prompt fidelity (CLIP score) and condition fidelity metrics, and between CLIP score and LPIPS. CLIP vs Canny IoU gives $r = 0.112$ and CLIP vs Circle IoU gives $r = 0.105$ ($r^2 < 0.02$ in both cases), confirming prompt fidelity and condition fidelity are effectively independent. CLIP vs LPIPS gives $r = -0.261$ ($r^2 = 0.068$), indicating prompt fidelity and condition fidelity are also only weakly linked. Together, these results motivate evaluating controllable generation along all three axes rather than any single one.

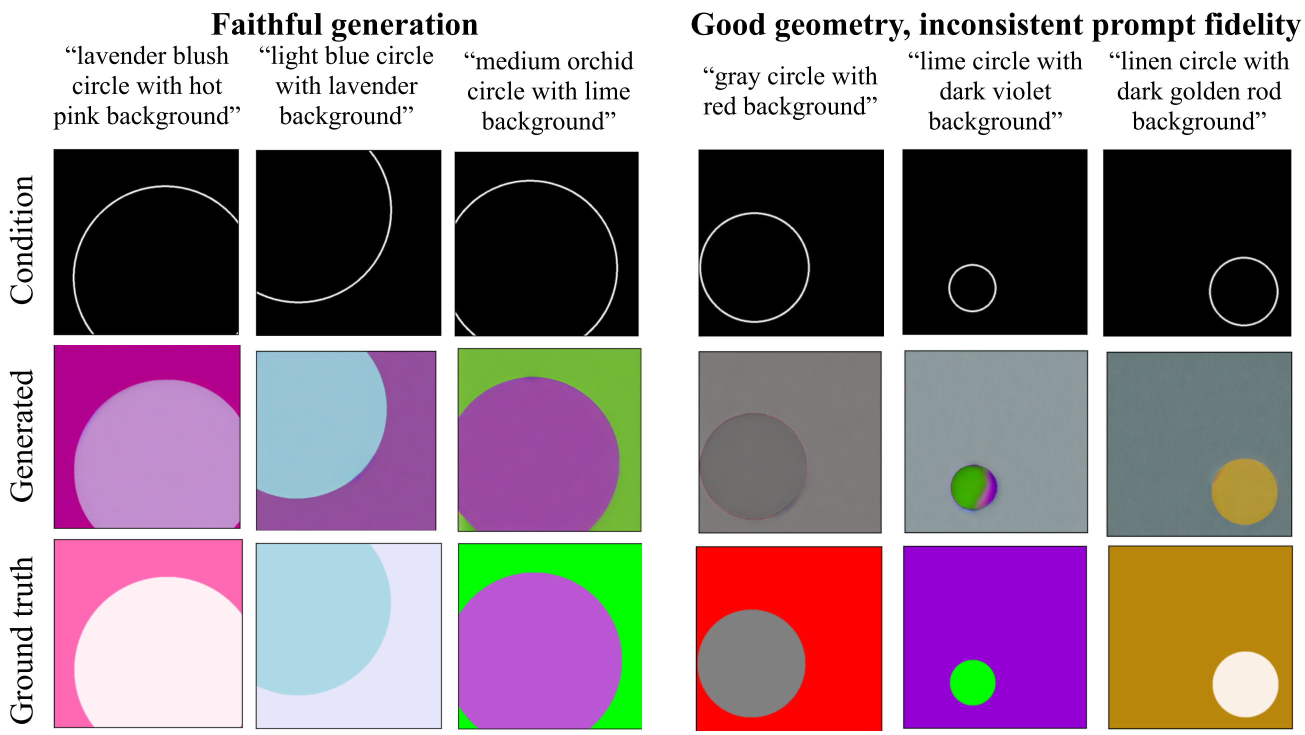

Figure 4. **ControlNet generates geometrically correct circles but with inconsistent prompt/color fidelity.** For the most part, generated images are consistently clean and well-rendered, and the model reliably reproduces circle geometry from the edge condition. However, ControlNet sometimes ignores color or appearance cues, producing plausible but prompt-misaligned outputs. When prompt fidelity degrades, generated images tend towards muted tones.

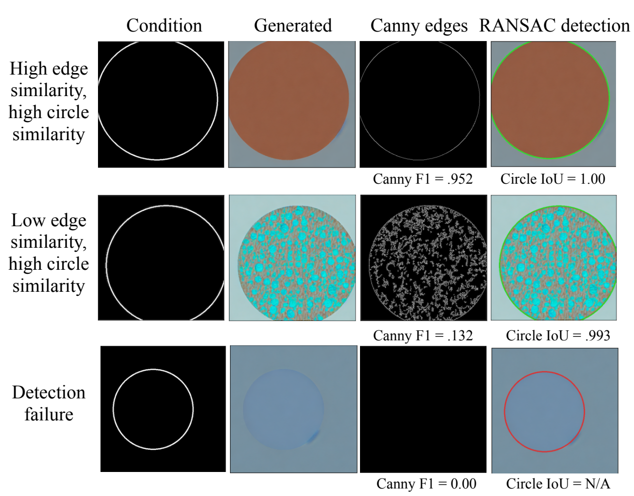

Figure 5. **Canny and RANSAC metrics capture different aspects of condition fidelity.** Row 1: both metrics indicate successful generation. Row 2: internal texture suppresses Canny IoU/F1 despite correct geometry. Row 3: Low contrast causes Canny edge extraction, and thus RANSAC detection, to fail.

Axis	Metric	Mean	Std
Image Quality	FID ↓	58.79	—
	LPIPS ↓	0.524	±0.220
Prompt Fidelity	CLIP ↑	30.26	±2.91
Condition Fidelity (Edges - Canny)	Canny IoU ↑	0.161	±0.086
	Canny F1 ↑	0.637	±0.385
	Chamfer ↓	19.80px	±30.36
Condition Fidelity (Geometry - RANSAC)	Detection Rate ↑	93.5%	—
	Circle IoU ↑	0.925	±0.157
	Center Error ↓	7.94px	±18.85
	Radius Error ↓	7.71px	±18.59

Table 1. Baseline results over 5,000 test images. Geometric metrics are computed over the 93.5% of images where RANSAC detection succeeded. High variance on center and radius error reflects RANSAC outliers. LPIPS computed per-image against ground truth. ↑ higher is better; ↓ lower is better.

3.3. Corruption Robustness Analysis

. We present a qualitative and quantitative assessment of the model's prompt fidelity, condition fidelity, image quality on each perturbation type and severity introduced. We treat corruption type and corruption severity as controlled ablation variables, varying one input property while hold-

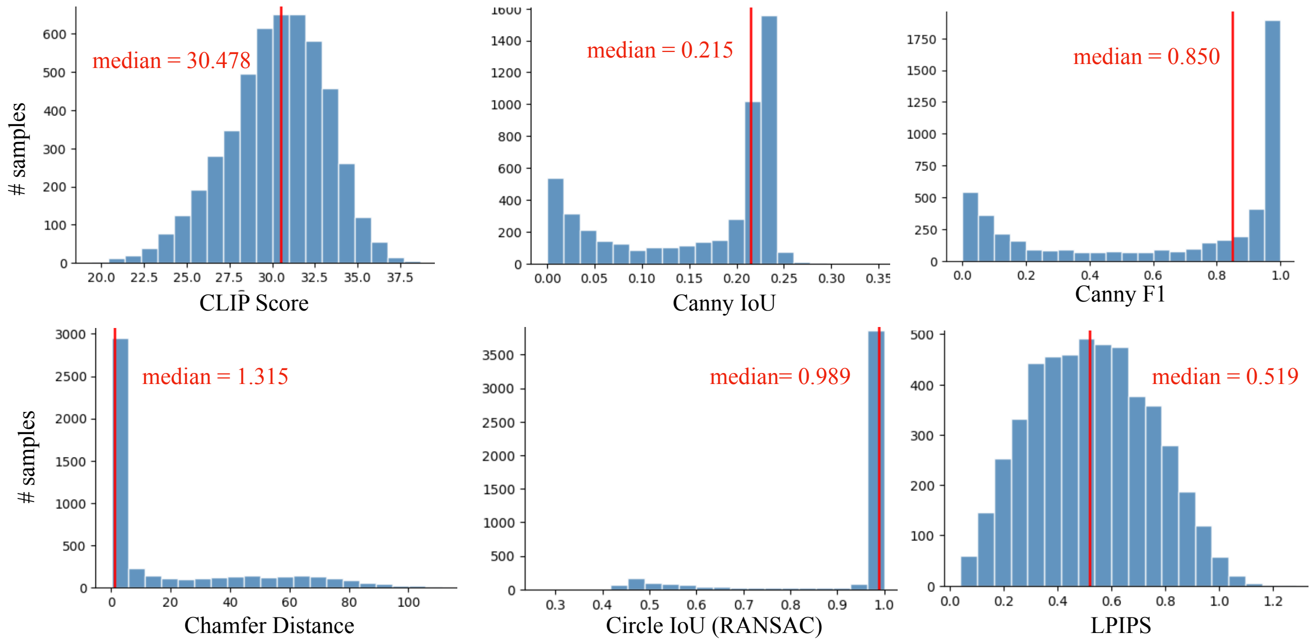

Figure 6. **Condition fidelity metrics are bimodal: the model either follows the condition or fails entirely.** Canny IoU and F1 show two distinct clusters with little middle ground. Circle IoU is heavily concentrated near 1.0 when detection succeeds. LPIPS (lower=better) shows a broad unimodal distribution, indicating variable perceptual quality across the test set.

ing the model, sampler, prompt, evaluation metrics, and test subset fixed. This allows direct comparison across corruption types at matched severities, and across severity levels within each corruption type.

Because the severity of one corruption type may not align proportionally to that of another, we evaluate metric degradation for each corruption type independently. Figure 7 shows example generated outputs at each severity, and Figure 8 shows metric degradation across all conditions relative to the clean baseline.

Overall, regardless of input corruption, we find that ControlNet is still capable of generating semantically correct and recognizable images. This is reflected by the fact that CLIP score is flat across for all corruptions and severities. LPIPS remains near the baseline for all corrupted images, indicating that the model is able to preserve perceptual image quality even if condition fidelity degrades.

Arc occlusion Figure 8 shows Canny F1 degrading the fastest and most severely, dropping to approximately 0.35 and 0.25 respectively by CS=0.6, while Circle IoU and Detection Rate remain above 0.8. Figure 7 shows that even at CS=0.6, the model generates a geometrically plausible circle despite the majority of the arc being removed, though edge appearance suffers significantly.

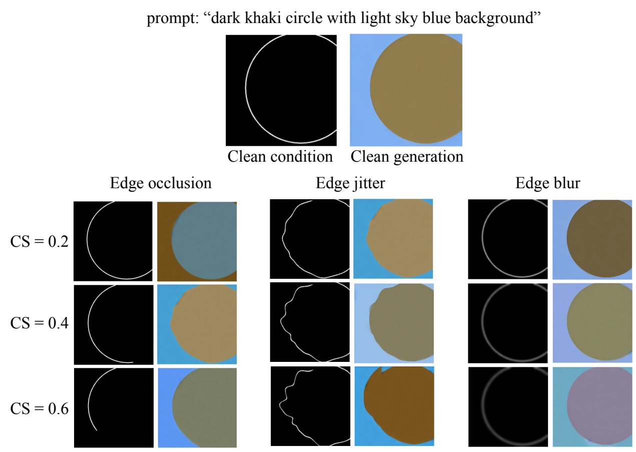

Figure 7. **Generated outputs under corrupted conditioning inputs for a representative test image.** Top: the clean conditioning input and corresponding generation. Bottom: corrupted conditioning inputs (left column of each pair) and generated outputs (right column) at three severity levels (CS = 0.2, 0.4, 0.6) across three corruption types. Under arc occlusion, the model continues to generate a geometrically plausible circle as the arc is progressively removed. Under edge jitter, generated circles become increasingly irregular at higher severities. Under Gaussian blur, generations remain close to the clean baseline throughout, with only subtle color shifts at the highest severity.

Edge jitter Edge jitter has a significant impact on edge metrics. Figure 8 shows Canny F1 collapsing more sharply

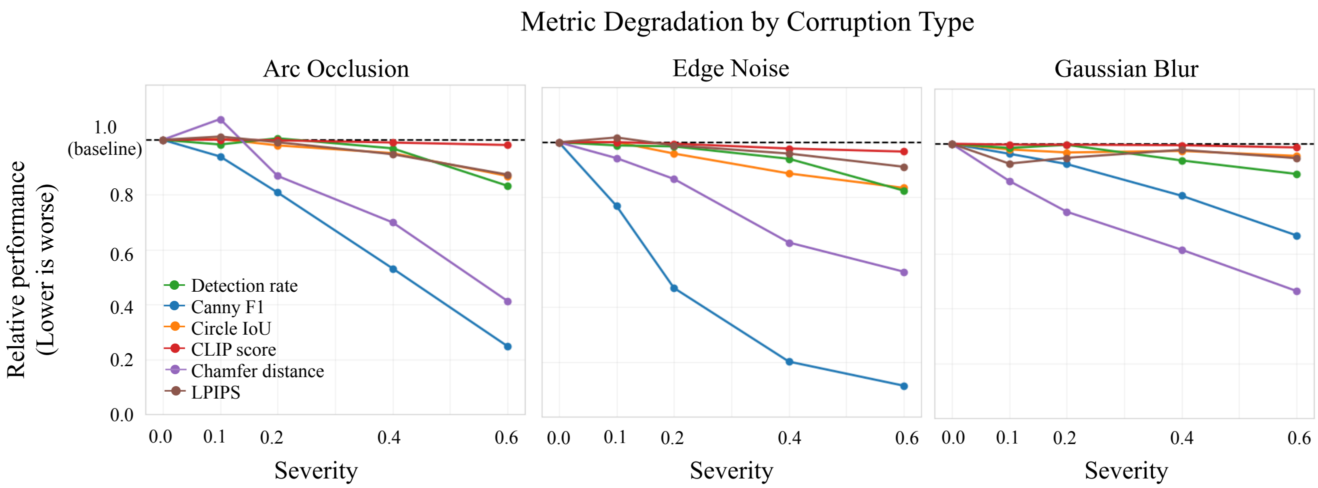

Figure 8. **ControlNet maintains prompt fidelity and perceptual quality under all corruptions, but condition fidelity degrades at different rates depending on corruption type.** Metrics are shown relative to the clean baseline (dashed line = 1.0). CLIP score and LPIPS remain near baseline across all conditions, while Canny-based metrics degrade most sharply under edge noise. Gaussian blur causes the least degradation overall, while arc occlusion produces steady decline in edge metrics with geometric metrics remaining more robust.

than under arc occlusion, dropping near zero by CS=0.4, with Chamfer distance also degrading significantly. This suggests the generated edges are not just noisy but spatially displaced as the model attempts to fit a smooth circular boundary onto a geometrically distorted input. Figure 7 shows generated circles becoming increasingly irregular at higher severities, mirroring distorted input contour. CLIP similarity remains flat throughout, which suggests the model falls back on the text prompt when the geometric signal is corrupted. This represents a soft failure mode: perceptual quality is maintained while geometric precision is lost, which is consistent with the fidelity-quality gap observed in the clean evaluation.

Gaussian blur Interestingly, edge blur has little to no impact on the fidelity of model generations. Figure 8 shows almost all metrics remaining near the clean baseline of 1.0, with the exception of Chamfer distance which degrades at CS=0.4–0.6. As shown in Figure 7, the model is able to accurately infer and generate the circle location even when the conditioning contour is heavily blurred at CS=0.6.

Overall, ControlNet maintains high prompt fidelity and perceptual quality across all corruption types and severities, with CLIP score and LPIPS remaining near the clean baseline throughout. Some degradation in edge fidelity occurs across all conditions, but geometric fidelity is largely preserved, as for most inputs, the model still correctly places and generates a recognizable circle.

Model performance degrades most severely under arc occlusion and least under Gaussian blur. This ordering seems to be consistent with the nature of each corruption, as blurring preserves all structural information in a spatially diffused form, while arc occlusion removes entire portions of the contour. We hypothesize that this asymmetry arises because ControlNet is trained to align image generation with observed conditioning features, not to perform analytic shape completion. Blurring weakens the contour but leaves spatial evidence distributed around the circle, whereas occlusion eliminates that evidence in some regions. The model therefore appears more robust to corruptions that degrade the signal but preserve its global support, than to corruptions that remove structural information entirely.

These findings have direct implications for how condition fidelity should be evaluated in controllable generation research. Prior work characterizes condition fidelity primarily through qualitative inspection on clean inputs, leaving two important questions unanswered: how precisely can condition fidelity be measured, and how does it degrade under realistic imperfect inputs? Our clean baseline results demonstrate that condition fidelity is precisely measurable using complementary geometric and edge-based metrics for contour circles, and that it is partially decoupled from both prompt fidelity and image quality. Our corruption study further shows that this decoupling persists and worsens under realistic input corruptions, with condition fidelity degrading while standard image quality and prompt fidelity metrics remain near baseline. Together, these findings suggest that current evaluation practices may systematically overestimate how well controllable generation systems follow their conditioning inputs, and that explicit condition fidelity evaluation is necessary to surface failures that image quality metrics alone cannot detect.

3.4. Limitations and Future Work

Several limitations of our current approach are worth noting. First, our evaluation is scoped to circle-contour conditions on Fill50K. While this simplicity enables precise geometric measurement, it limits generalizability to more complex conditions such as human poses or depth maps, where condition fidelity is harder to quantify and corruption effects may differ substantially. Extending this corruption study to richer condition types is an important future direction.

Second, our RANSAC-based geometric evaluation has known failure modes. When the generated circle extends significantly beyond the image boundary, or when boundary contrast is very low, RANSAC may fail to detect a circle even when one is present. The 6.5% non-detection rate reflects a mix of model failures and detector limitations that are difficult to distinguish. Improved geometric fitting methods could address this ambiguity.

Third, we were able to achieve satisfactory generations with limited compute by training the ControlNet Fill50k for just two training epochs. However, even though Fill50K is a very aligned dataset, our model occasionally generates extraneous textures or fails to exactly follow both the input text and condition. We qualitatively review common generation-related failure modes exhibited by our model in Figure 9 and Section 3.4. We hypothesize that using a larger batch size and additional training steps will further improve the alignment of the model generations.

Finally, region-specific color accuracy is not explicitly captured by any of our current metrics. CLIP score measures overall semantic alignment, but may not explicitly distinguish whether the correct color was assigned to the foreground versus background. A dedicated color accuracy metric would strengthen the prompt fidelity evaluation axis.

Failure Cases We qualitatively discuss several recurring failure modes revealed by our evaluation.

First, generated images with heavy internal texture produce noisy Canny edge maps that score poorly against the clean condition contour even when the circle geometry is correct, as the model follows the condition geometrically but generates visually complex artifacts that confuse edge detection (Figure 5, Row 2). Using a more texture-robust edge detector or applying morphological filtering before metric computation could mitigate this.

Second, low-contrast generations cause Canny to detect no boundary at all, giving IoU and F1 of zero despite a visually present circle (Figure 5, Row 3), likely occurring when the prompt color is close in value to the background. Contrast-aware post-processing or learned edge detection could address this.

Third, foreground and background colors are occasionally swapped or blended, indicating that the semantic relationship between prompt attributes and spatial regions has not been fully learned (Figure 9). This is possibly due to limited training steps, and longer training or explicit color-region binding losses could reduce this.

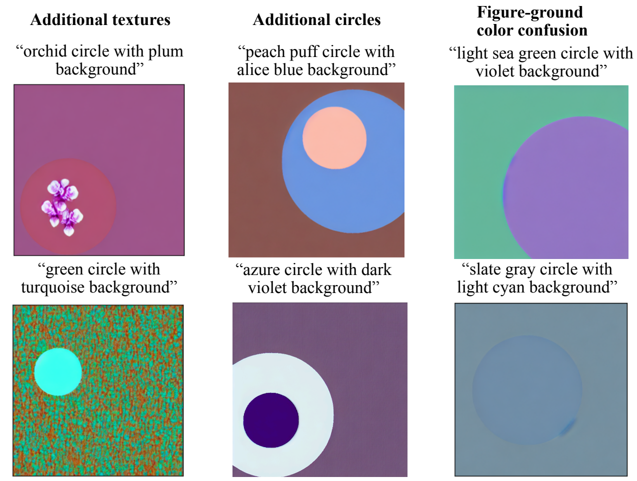

Figure 9. **Common generation-related failures.** Column 1: ControlNet generates extraneous textures, often for prompts with color names containing objects, like "orchid" or "turquoise." Column 2: ControlNet generates multiple nested circles. Column 3: ControlNet swaps the intended foreground and background colors, or forgets about one of the colors entirely.

Fourth, prompts with referential names (e.g. "lilac", "blanched almond") anecdotally tend to produce textured outputs, suggesting the model conflates color semantics with visual textures (Figure 9).

More broadly, having characterized how ControlNet fails under corrupted inputs, an interesting next step is to investigate whether training with augmented or corrupted conditions could improve robustness. Our findings suggest that the ControlNet encoder has learned pattern matching rather than abstract geometric rules, which points toward data augmentation during training as a promising mitigation strategy.

3.5. Ethical and Societal Considerations

Our experiments use a synthetic dataset containing no people, demographic attributes, or sensitive information, and focus on model fidelity and evaluation rather than content generation, limiting direct risks of bias or harm.

More broadly, however, controllable T2I models raise important ethical and societal concerns. Robust control in models can be beneficial by allowing users to generate intended outputs more reliably, but it can also enable more precise image generation in misleading or sensitive contexts. Failures in condition following can also matter, *e.g.*, a user may overtrust a visually plausible image even when it does not satisfy the requested condition, which is especially concerning in high-stakes applications. These concerns motivate several mitigation strategies, including transparent reporting of condition fidelity alongside image quality metrics (as we advocate in this work), development of standardized multi-axis evaluation benchmarks to hold models accountable beyond perceptual quality, and deployment-time guardrails, such as automatic condition fidelity verification before surfacing outputs to users.

4. Conclusion

This paper investigates how robustly a trained ControlNet preserves condition fidelity under corrupted conditioning inputs, addressing a gap in existing evaluation practices for controllable text-to-image generation. We train a ControlNet on Fill50K and introduce a rigorous multi-axis evaluation framework spanning prompt fidelity, image quality, and condition fidelity. A core contribution is our condition fidelity evaluation: two complementary metrics, Canny-based edge comparison and RANSAC-based geometric estimation, that together capture distinct failure modes beyond the single-extractor approach of prior work. We further conducted a controlled corruption study evaluating three perturbation types at four severity levels.

Our clean baseline results show that ControlNet reliably generates geometrically correct circles, and our evaluation framework yields a 93.5% circle detection rate and average Circle IoU of 92.5% in generated images, while prompt fidelity and image quality metrics confirm consistent semantic alignment. Under corruption, prompt fidelity and perceptual quality remain stable across all conditions while condition fidelity degrades at rates that depend strongly on corruption type, with edge jitter causing the sharpest collapse and Gaussian blur causing the least. Together, these results demonstrate a *fidelity-quality gap*: standard image quality metrics fail to detect condition fidelity failures, meaning a model can appear to succeed while failing at its core conditioning task.

These findings suggest that condition fidelity should be evaluated explicitly and separately from image quality in controllable generation systems, particularly as such models are deployed in settings where conditioning inputs may be noisy or user-provided. Our aim is for this work to encourage more systematic robustness evaluations as a standard component of assessing controllable generation.

5. AI Usage

We used ChatGPT and Claude to help brainstorm some of the ControlNet-related ideas mentioned in this paper and fill in conceptual gaps in our understanding, as well as help with some of the code implementation of extra evaluation and analysis frameworks and metrics. We did not use any Gen AI tools to write or revise the content of this document.

References

[1] John Canny. A computational approach to edge detection. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 8(6):679–698, 1986. 3

[2] Xin Chen, Gillian Dobbie, Xinyu Wang, Feng Liu, Di Wang, and Jingfeng Zhang. Robust learning of diffusion models with extremely noisy conditions, 2025. 2

[3] Martin A Fischler and Robert C Bolles. Random sample consensus: A paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981. 4

[4] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in Neural Information Processing Systems*, 2017. 3

[5] Ming Li, Taojiannan Yang, Huafeng Kuang, Jie Wu, Zhaoning Wang, Xuefeng Xiao, and Chen Chen. Controlnet++: Improving conditional controls with efficient consistency feedback. In *European Conference on Computer Vision*, 2024. 1, 2, 3, 4

[6] Chong Mou, Xintao Wang, Liangbin Xie, Jian Zhang, Zhongang Qi, Ying Shan, and Xiaohu Qie. T2i-adapter: Learning adapters to dig out more controllable ability for text-to-image diffusion models. *arXiv preprint arXiv:2302.08453*, 2023. 1, 2

[7] Can Qin, Shu Zhang, Ning Yu, Yilun Feng, Xinchao Yang, Yufei Zhou, Hao Wang, Juan Carlos Niebles, Caiming Xiong, and Silvio Savarese. Unicontrol: A unified diffusion model for controllable visual generation in the wild. In *Advances in Neural Information Processing Systems*, 2023. 1, 2

[8] Alec Radford et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021. 3

[9] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022. 1, 3

[10] Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *International Conference on Learning Representations*, 2021. 3

[11] Wenjie Xuan, Yufei Xu, Shanshan Zhao, Chaoyue Wang, Juhua Liu, Bo Du, and Dacheng Tao. When controlnet meets inexplicit masks: A case study of controlnet on its contour-following ability. In *Proceedings of the 32nd ACM International Conference on Multimedia*, pages 6979–6988, 2024. 2

[12] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models. In *IEEE/CVF International Conference on Computer Vision*, pages 3836–3847, 2023. 1, 2, 3, 4, 5

[13] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep

features as a perceptual metric. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 586–595, 2018. 3